PREDICTION OF FOOTBALL TRANSFER VALUES FOR THE 2025/26 SEASON: A DATA-DRIVEN APPROACH FROM EUROPE'S TOP FIVE LEAGUE

Nguyễn Huỳnh Minh Phú, Hà Xuân Hoàng

Nguyễn Gia Tuấn Anh, Trần Quốc Khánh

Trường Đại học Công nghệ Thông tin, Thành phố Hồ Chí Minh, Việt Nam

Đại học Quốc gia, Thành phố Hồ Chí Minh, Việt Nam

{23521183, 23520516} @gm.uit.edu.vn

Abstract

In the era of football commercialization, transfer value has emerged as a crucial consideration, playing a decisive role in shaping the policies and strategic direction of leading football clubs around the world. This study aims to analyze the key factors influencing a player's market value and to propose appropriate price predictions to support transfer decisions in the upcoming season. To facilitate this research, we developed a dedicated dataset titled *"Top 5 Leagues Football Market Transfer 2024/25"* for analysis and evaluation purposes.

The research process involved collecting player-related data, followed by data preprocessing to identify the most influential variables affecting market value. Subsequently, machine learning models were employed to predict player transfer values for the 2025/26 season.

By leveraging the most up-to-date market values from the 2024/25 season across major UEFA leagues, the model takes into account several key factors—such as age, number of goals, assists, and the league in which the player competes. The study offers valuable insights for football industry stakeholders, contributing to more effective player management strategies and data-driven navigation of the transfer market.

Keywords: transfer value, football, player, UEFA, machine learning

1 Introduction

In recent years, there has been a growing interest in the transfer market value of football players. This value not only reflects a player's individual talent and success but also significantly influences club policies, revenues, and the overall development potential of both clubs and leagues.

Against this backdrop, we conducted a project to construct and analyze football market values for the 2024/25 season across the top five European leagues. The study explores key factors directly impacting market value and provides predictive insights into a player's value for the upcoming 2025/26 season.

Currently, there are two main approaches to player valuation that researchers can adopt. The first is based on actual transfer fees paid by clubs at specific points in time. However, such figures are rarely disclosed officially and are often only partially revealed through specialized media outlets or football websites. The second approach relies on estimated market values provided by media organizations such as Transfermarkt.de, Marca, or Kicker.de—this latter method serves as the primary data source for our research.

Finally, we preprocessed the dataset and developed machine learning models to generate accurate predictions of player transfer values, thereby offering practical support to clubs in the upcoming transfer window.

Input: Valuable statistical indicators of a football player.

Output: Predicted market value of the player for the 2025/26 season.

2 Related Work

Numerous studies have been conducted in the field of football, particularly focusing on predicting players' transfer values.

Poli, R., Besson, R., and Ravenel, L. (2024) developed a Multiple Linear Regression (MLR) model incorporating variables such as age,

playing position, and minutes played. Their model successfully explained over 84% of the variance in transfer fees based on global data.

In contrast, Huang, C., and Zhang, S. (2023) applied a Gradient Boosting Decision Tree model combined with SHAP to assess the influence of each feature. Their findings identified technical skills, physical condition, and intelligence as key factors, with the model achieving an R^2 of up to 0.901 and an RMSE of 3.22 million.

Additionally, Lee, H., Tama, B. A., and Cha, M. (2022) employed a LightGBM model optimized via the Tree-structured Parzen Estimator (TPE) algorithm and interpreted with SHAP to predict player values. This model demonstrated superior performance compared to traditional regression models, reducing RMSE by a factor of 1.4 to 3.8.

These studies not only demonstrate the effectiveness of machine learning algorithms in forecasting player transfer values but also highlight the critical importance of selecting and analyzing appropriate features.

3 Dataset

To support this study, we collected data from two reputable sources: Transfermarkt and WhoScored – widely recognized platforms that provide reliable statistics on professional football players' performance and market value. These sources have also been trusted by numerous previous studies, including the works of Yalçınkaya, M. A., & Işık, M. (2024), Baouan, A., Bismuth, E., Bohbot, A., Coustou, S., Lacome, M., & Rosenbaum, M. (2022), and Ćwiklinski, B., Giełczyk, A., & Choraś, M. (2021).

The dataset comprises information on over 2,000 players currently competing in the top five European club leagues – *Premier League, La Liga, Serie A, Bundesliga, and Ligue 1* – during the 2024/25 season. In total, we collected **70 features**, categorized into four key aspects: *Basic player information* (e.g., Player, Club, League, Age, Height, etc.), *Attacking statistics* (e.g., SpG, TotGs, TotSh, TotDrib, etc.), *Defensive statistics* (e.g., TotTkl, DribPast, Fouled, TotSav, TotClr, etc.), *Passing statistics* (e.g., TotSP, AccLB, TotKPs, TotAss, etc.).

4 Experiment

4.1 Data collection

The data for this study were collected through web scraping from two websites: Transfermarkt and WhoScored. Specifically, using Python's Selenium library, we implemented automated web scraping to extract relevant features.

From Transfermarkt, we gathered player market values, profile information (such as age, nationality, height, foot, position, and club), as well as the year a player's contract ends. Meanwhile, WhoScored provided detailed performance statistics from the 2024/25 season, including offensive metrics (goals, shots, dribbles, key passes), defensive metrics (tackles, interceptions, clearances) and passing indicators (number of passes, accuracy, through balls). The final dataset comprises approximately 70 features per player, covering players from the top five European leagues during the 2024/25 season.

After scraping, the extracted raw data from multiple sources were then consolidated and merged into a complete dataset. A preliminary statistical analysis revealed that the dataset consists of 10 categorical features (presented in *Table 1*) and 60 numerical features.

Attribute	Description
Age	Age and Date of Birth
Height	Height (cm & m)
National	Nationality
League	One of top 5 UEFA European leagues
Club	Club the player is currently playing for.
Main_position	Player's primary playing position.
Foot	Player's dominant foot.
Market_Value	2024/25 Transfer Value
Contract	The year a player's contract ends.
Apps	Number of matches started and as a substitute.

Table 1. The list of ten categorical features

4.2 Data Preprocessing

The dataset reflects a wide range of player market values, spanning from €25,000 to €200 million. *Figure 1* illustrates the distribution of these player values.

Upon examining the distribution in *Figure 1*, we observed a significant skew toward lower value ranges. To address this skewness and enhance the stability and learnability of the data for machine learning models, we applied a log transformation, specifically $\log(1 + p)$.

This transformation substantially improved the symmetry of the data distribution, as illustrated in *Figure 2*.

4.3 Format Normalization

After the data collection process, we identified several features with inconsistent or non-standard formats that could negatively impact model performance. For example, the Height attribute included unnecessary text such as ‘cm’ or ‘m’, and the Age feature combined both the player's age and full date of birth.

To address these issues, we performed data cleaning and normalization, standardizing the formats and converting categorical features into numerical representations to ensure compatibility with machine learning algorithms.

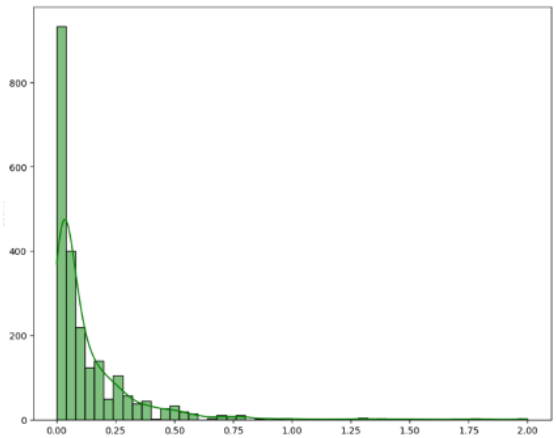

Figure 1. The distribution of Market Value before transformation

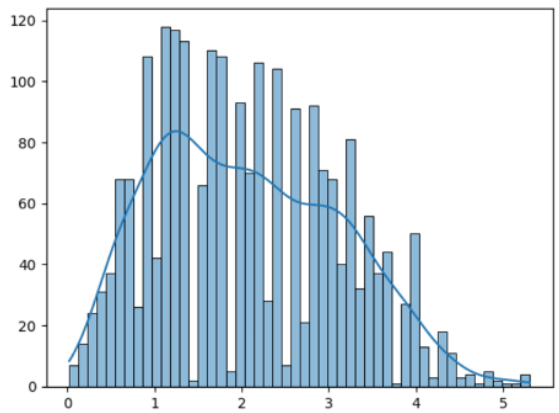

Figure 2. The distribution of Market Value after transformation

Feature	Before	After
Age	21 years old (05-01-2004)	21
Height	1,85m	185
Market_Value	€2.80m	2800000
	€600k	2800000

Table 2. Status before and after transfering

4.4 Feature Extraction

In addition to format standardization, the study also performed feature extraction and transformation to improve model performance. Specifically, for the *Contract* feature, we calculated the remaining contract duration (in months at the time of data collection) and converted it into a new feature called *Month_Left.*

The original *Nation* feature included over 90 countries, which made standardization difficult and posed a risk of degrading model performance. Therefore, we grouped countries by their respective continents, resulting in a new feature called *Continent*.

Similarly, the *Main_Position* feature was simplified into broader position categories (*Position_Group*) to reduce classification complexity. The *YelC* and *RedC* features were also combined into a composite feature named *Card*. In addition, the *Club* feature was grouped into *Club_Group* based on club level and regional similarity.

Before	After	Detail
Contract	Months_left	Jun 30, 2027 → 24
Apps	Start_apps & Sub_app	0(7) → 0 & 7
Red + Yel	Card	5 Red + 1 Yel → 11 Cards
National	Continent	France → Europe
		Japan → Asia
Main_Position	Position_Group	Centre-Forward → Tiền Đạo
		Goalkeeper → Phòng Ngự
Club	Club_Group	Read Marid → Top Tier
		Man City → Top Tier

Table 3. Feature Extraction Before and After

4.5 One-Hot Encode

After feature extraction, the study applied One-Hot Encoding to categorical features to convert them into numerical format and ensure compatibility with machine learning models.

The following categorical features were transformed using One-Hot Encoding to ensure compatibility with machine learning models: *Foot, League, Position_Group, Continent,* and *Club_Group.*

4.6 Feature Selection

One of the key steps in the analysis process is feature selection. With a dataset comprising 70 features, the risk of multicollinearity may negatively impact the accuracy of prediction models. Therefore, the study employed correlation matrix analysis (*Figures 3 and 4*) and the Variance Inflation Factor (VIF) (*Table 4*) to eliminate redundant or highly correlated features.

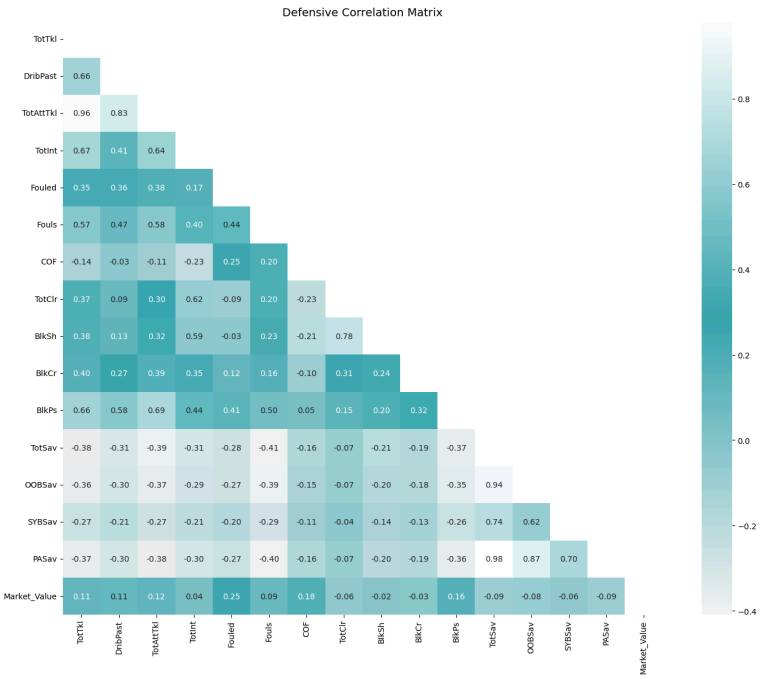

Figure 3. Correlation matrix of Offensive feature

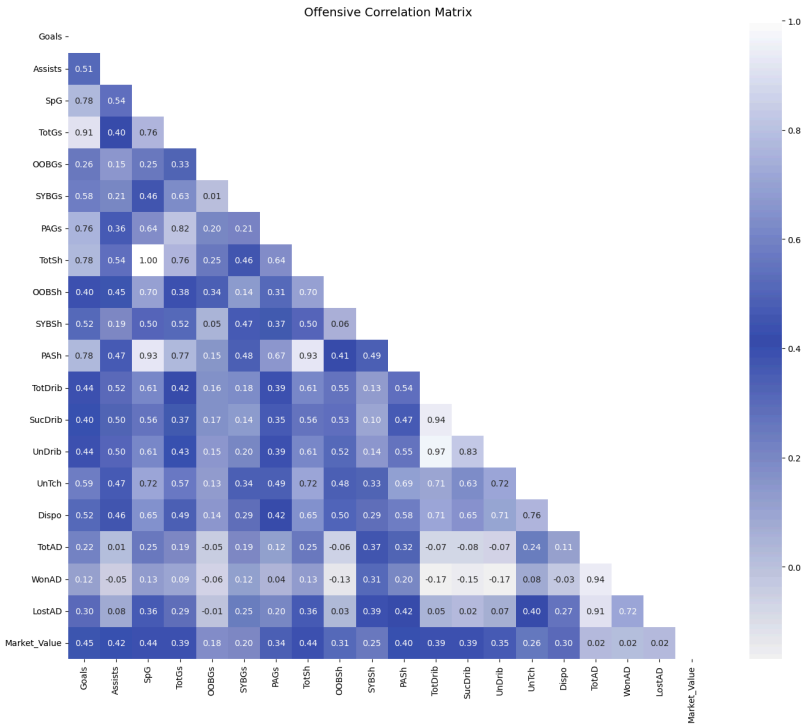

Figure 4. Correlation matrix of Offensive features

Feature	VIF
TotSh	inf
WonAD	inf
SpG	inf
AerialsWon	inf
TotPs	72462.68
AccSP	55632.07
TotSav	3040.07
InAccLB	1741.30
⋮	⋮
CorAss	1.41
ThrbAss	1.39
RedC	1.14
FreAss	1.10

Table 4. Variance Inflation Factor of selected features

According to the statistics in *Table 4*, all features with a VIF value exceeding the threshold of 10 were removed, except for Goals and Mins, which were retained due to their practical significance in determining a player's market value.

After the preliminary screening process, we selected *48 features* that have a positive impact on the model's output.

4.7 Remove Outliers

Finally, we removed data samples with outlier transfer values. Two methods were tested: *Interquartile Range (IQR)* and *Winsorizing.*

IQR (Interquartile Range) is a statistical measure of data dispersion commonly used to detect and handle outliers in a dataset. In this case, we removed outliers using the IQR method.

$$\mathbf{IQR = Q3 - Q1}$$

where Q1 represents the first quartile (25th percentile), the value below which 25% of the data fall, while Q3 is the third quartile (75th percentile), below which 75% of the data lie. The IQR captures the middle 50% of the dataset, representing the range between Q1 and Q3.

In contrast, **Winsorizing** does not remove data points but instead **replaces extreme values** (typically the lowest and highest 5% or 10%) with the closest values within a specified percentile range, thereby reducing the influence of outliers without discarding any samples.

Based on the model performance evaluation results, **the IQR method** was preferred due to its greater stability and higher accuracy.

4.8 Jarque-Bera Test

The Jarque–Bera (JB) test is a widely used statistical method for testing whether a variable follows a normal distribution. This test specifically relies on two characteristics of the distribution: skewness and kurtosis. In the context of machine learning, the JB test is often applied to assess the normality assumption of residuals in regression models. A large JB statistic and a p-value less than 0.05 indicate that the residuals are unlikely to follow a normal distribution, which may affect the validity of subsequent statistical inferences or the assumptions underlying linear models.

In this study, the JB test was conducted on all input features. The results show that *all variables yielded p-values < 0.01*, suggesting that they significantly deviate from a normal distribution. Therefore, the assumption of normality does not hold for these features.

Feature	JB Test (p-value)
Age	55.050 (<0.01)
Height	17.369 (<0.01)
Market_Value	54917.640 (<0.01)
Months_left	83.942 (<0.01)
Start_app	158.856 (<0.01)
⋮	⋮
Club_Group_Lower_Tier	714.275 (<0.01)
Club_Group_Mid_High Tier	2911.199 (<0.01)
Club_Group_Mid Tier	961.137 (<0.01)
Club_Group_Top Tier	8734.711 (<0.01)
Club_Group_Upper Tier	6559.929 (<0.01)

Table 5. Jarque-Bera Test's result of ten features

4.9 Machine Learning

This section presents a description of the research methods employed in this study. The primary objective is to identify the most effective model for generating reasonable predictions of player transfer values. To achieve this, we trained and evaluated five widely-used regression models: *Ridge Regression, Random Forest Regression, Support Vector Regression (SVR), LightGBM Regression,* and *XGBoost Regression.*

4.9.1 Model Development and Optimization

During the model development process, the dataset was randomly split into three subsets: 60% for training, 20% for validation, and the remaining 20% for testing purposes.

Processing Step	Total Samples	Train (60%)	Dev (20%)	Test (20%)
Original Dataset	2249	1349	450	450
After IQR Filtering	2067	1240	413	414
After Winsorizing	2249	1349	450	450

Table 6. Sample Size and Dataset Splitting at Each Preprocessing Stage

To optimize the hyperparameters of ensemble learning models, we employed cross-validation in conjunction with grid search.

Ultimately, the final model was constructed by leveraging the selected features and fine-tuning hyperparameters to achieve optimal performance.

4.9.2 Ridge Regression

Ridge Regression is a variant of linear regression designed to mitigate multicollinearity and prevent overfitting. This approach incorporates an L2 penalty term into the loss function, which shrinks the regression coefficients and contributes to greater model stability.

The most critical hyperparameter in Ridge Regression is the regularization strength (commonly denoted as *alpha*), which controls the degree of penalization. By applying Grid Search in conjunction with cross-validation, the optimal value for *alpha* was determined to be 0.001.

4.9.3 Support Vector Regression

Support Vector Regression (SVR) is a regression technique based on the Support Vector Machine (SVM) algorithm, which aims to identify an optimal hyperplane that minimizes the prediction error within a specified margin of tolerance (ε-insensitive zone). SVR is particularly effective in modeling complex and nonlinear relationships by utilizing kernel functions, thereby enabling the model to capture intricate patterns in the data.

4.9.4 Random Forest

Random Forest Regression (RFR) is an ensemble machine learning method that combines multiple decision trees to enhance prediction accuracy and generalization. Each tree is trained on a random subset of the data, and the final prediction is obtained by averaging the outputs of all trees. This approach mitigates the risk of overfitting commonly associated with individual decision trees and performs well on nonlinear, noisy, and high-dimensional datasets.

4.9.5 XGBoost

XGBoost Regression is a gradient boosting-based regression algorithm that builds an ensemble of decision trees sequentially, where each new tree focuses on correcting the residuals of the previous ensemble. It incorporates both L1 and L2 regularization to control overfitting, while also optimizing computational efficiency in terms of speed and memory usage. Due to its robustness with high-dimensional and nonlinear data, XGBoost is well-suited for predicting player transfer values based on both statistical and personal attributes.

4.9.6 LightGBM

LightGBM Regressor is a gradient boosting-based regression algorithm developed by Microsoft, designed for high training speed and scalability to large datasets. Unlike traditional boosting methods, it adopts a leaf-wise tree growth strategy and histogram-based learning, which enhances accuracy and reduces training time. Its efficiency on high-dimensional, imbalanced, and nonlinear data makes it well-suited for predicting football player transfer values in this multivariate regression task.

5 Result

5.1 Metrics

To evaluate and compare the performance of the models, we employed five evaluation metrics**:** *Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE),* and *R-squared (R^2).*

5.1.1 Mean Squared Error (MSE)

Mean Squared Error (MSE) is a commonly used metric to measure the average squared difference between predicted and actual values. A lower MSE indicates better model performance. The formula for MSE is:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where y_i denotes the actual value, $\widehat{y_i}$ the predicted value, and n is the number of observations.

5.1.2 Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) is the square root of the Mean Squared Error and represents the average deviation between predicted and actual values, measured in the original units of the target variable. RMSE is widely used for evaluating the accuracy of regression models due to its interpretability. The formula is:

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

5.1.3 R^2 Score

R^2 Score (Coefficient of Determination) is a statistical measure used to evaluate the goodness-of-fit of a regression model. It represents the proportion of the variance in the dependent variable that is explained by the independent variables in the model. Formally, it is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

An R^2 value close to 1 indicates that the model explains a large portion of the variance in the response variable, while a value near 0 suggests poor explanatory power. Negative values imply that the model performs worse than a naive baseline that always predicts the mean.

5.1.4 Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error (MAPE) is employed to assess the prediction accuracy of the model. It measures the average absolute percentage difference between the predicted values and the actual values. The metric is particularly intuitive as it expresses error as a percentage, allowing for easier interpretation regardless of the data scale. The MAPE is computed as follows:

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\%$$

5.1.5 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) is a commonly used metric to evaluate the accuracy of regression models. It measures the average magnitude of errors between predicted and actual values, without considering their direction. Unlike other metrics that square the errors, MAE uses the absolute value, making it straightforward and interpretable:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

5.2 Results

The performance results obtained after applying Ridge Regression, Support Vector Regression (SVR), Random Forest (RF), XGBoost, and LightGBM models on the training, development, and test datasets are summarized in *Table 6*.

Model	Dataset	MSE	RMSE	R2	MAE
RG	Train	3.72	1.93	0.95	1.41
	Dev	3.61	1.90	0.95	1.41
	Test	3.35	1.83	0.95	1.35
SVR	Train	0.63	0.79	0.99	0.75
	Dev	7.94	2.82	0.90	2.04
	Test	7.50	2.74	0.89	1.95
RF	Train	2.93	1.71	0.96	1.30
	Dev	4.01	2.00	0.95	1.44
	Test	3.25	1.80	0.95	1.31
XGB	Train	2.36	1.53	0.97	1.14
	Dev	3.88	1.97	0.95	1.42
	Test	3.50	1.87	0.95	1.34
LGBM	Train	13.44	3.67	0.83	2.59
	Dev	27.19	5.21	0.66	3.72
	Test	26.07	5.11	0.63	3.72

Table 7. Results

5.3 Durbin-Watson Test

The Durbin–Watson test is a widely used diagnostic in linear regression, aimed at detecting the presence of autocorrelation in the residuals.

After training the Ridge Regression model, we applied the Durbin – Watson test to evaluate the independence of residuals. The test yielded a statistic of **2.089**, which is close to the ideal value of 2. This indicates no significant evidence of first-order autocorrelation, suggesting that the assumption of independent observations holds.

6 Discussion

The objective of this study is to consolidate machine learning models to identify and analyze the most influential factors affecting athletic performance. During the research process, we experimented with various data transformation techniques, including Box-Cox, Yeo-Johnson, and log(1+p), to determine the most effective approach for optimizing model performance.

While our study offers valuable insights, it also has several limitations. First, the current models are unable to accurately estimate the market value of superstar players, as this requires consideration of numerous external factors beyond on-field performance metrics. Second, our dataset was compiled from publicly available sources, which inevitably leads to discrepancies between the provided information and actual confidential data. This limitation constrains the depth and scope of the conclusions that can be drawn from our analysis.

7 Conclusion

Based on our research findings, we draw several key conclusions to inform the development of market value prediction models.

The data preprocessing phase plays a critical role and is a determining factor in the model's overall accuracy.

By leveraging a wide range of player attributes and position-specific features, the model provides reliable estimates of market value, supporting football clubs in making informed decisions regarding transfers and financial planning.

This research not only promotes the application of data-driven techniques in sports economics but also establishes a flexible framework that can be adapted to various other domains where objective and data-rich valuation is essential.

As machine learning models become more sophisticated and datasets more comprehensive, the potential for accurate, automated valuation in both sports and other fields will continue to expand—offering valuable insights for practical applications and academic research alike.

8 References

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10. Phụ lục - Appendices

A. FEATURE DESCRIPTION

Football_Player_Profile Dataset			
No.	**Attributes**	**Type**	**Description**
1	Player	String	Tên cầu thủ
2	Club	String	Câu lạc bộ đang thi đấu
3	League	String	Giải đang thi đấu
4	Age	String	Tuổi và ngày tháng năm sinh
5	Height	String	Chiều cao
6	National	String	Quốc tịch
7	Positions	String	Các vị trí sở trường
8	Main_Position	String	Vị trí sở trường chính
9	Foot	String	Chân thuận
10	Apps	Int	Số lần ra sân
11	Mins	Float	Tổng số phút ra sân
12	Goals	Float	Số lượng bàn thắng
13	Assist	Float	Số kiến tạo
14	Yellow Card - YelC	Float	Số thẻ vàng
15	Red Card - RedC	Float	Số thẻ đỏ
16	Pass Success Percentage - PS%	Float	Tỷ lệ thành công các đường chuyền
17	MotM	Float	Số lần được chọn là cầu thủ xuất sắc nhất trận
18	Rating	Float	Điểm trung bình được các nhà báo thể thao đánh giá sau mỗi trận đấu
19	Link	Float	Link profile cầu thủ trên trang TransferMarket
20	Remaining Contract Year - RCY	Float	Số năm còn lại của hợp đồng
21	Value	Float	Giá trị chuyển nhượng trên thị trường mùa giải 2024 - 2025

Offensive_Profile Dataset			
No.	**Attributes**	**Type**	**Description**
1	Shots per Game - SpG	Float	Số cú sút trung bình
Goals - Nhóm dữ liệu về bàn thắng			
2	Total Goals - TotGs	Float	Đây là tổng số bàn thắng được ghi trong trận đấu, bao gồm tất cả các bàn thắng từ mọi khu vực trên sân.
3	OutOfBox Goals - OOBGs	Float	Số bàn thắng được ghi từ ngoài khu vực vòng cấm. (ngoài phạm vi 16,5 mét từ khung thành)
4	SixYardBox Goals - SYBGs	Float	Số bàn thắng được ghi trong khu vực 6 yard (khoảng 5,5 mét) gần khung thành.
5	PenaltyArea Goals - PAGs	Float	ô bàn thắng được ghi trong khu vực vòng cấm (16,5 mét từ khung thành).
Shots - Nhóm dữ liệu về các cú sút			
6	Total Shots - TotSh	Float	Tổng số cú sút của đối phương trong suốt trận đấu, bao gồm tất cả các pha sút từ mọi khu vực trên sân.
7	OutOfBox Shots - OOBSh	Float	Số cú sút thực hiện từ ngoài khu vực vòng cấm
8	SixYardBox Shots - SYBSh	Float	Số cú sút thực hiện trong khu vực 6 yard (khoảng 5,5 mét) gần khung thành
9	PenaltyArea Shots - PASh	Float	Số cú sút thực hiện trong khu vực vòng cấm
Dribbles - Nhóm dữ liệu về thông số dẫn bóng			
10	Total Dribbles - TotDrib	Float	(Tổng số lần dẫn bóng): Đây là tổng số lần cầu thủ thực hiện các pha dẫn bóng trong suốt trận đấu
11	Unsuccessful Dribbles - UnDrib	Float	(Dẫn bóng không thành công): Đây là số lần cầu thủ cố gắng dẫn bóng nhưng bị đối phương chặn lại hoặc mất bóng
12	Successful Dribbles - SucDrib	Float	(Dẫn bóng thành công): Đây là số lần cầu thủ dẫn bóng qua đối phương một cách thành công,
Possession loss - Nhóm dữ liệu về thông số chạm bóng			
13	Unsuccessful Touches - UnTch	Float	(Chạm bóng không thành công): Đây là số lần cầu thủ chạm vào bóng nhưng không tạo ra tình huống tích cực
14	Dispossessed - Dispo	Float	Đây là số lần cầu thủ bị đối phương lấy bóng từ chân mình
Aerial - Nhóm dữ liệu về thông số không chiến			
15	Total Aerial Duels - TotAD	Float	(Tổng số tình huống không chiến): Đây là tổng số lần cầu thủ tham gia vào các tình huống tranh chấp bóng trên không
16	Won Aerial Duels - WonAD	Float	(Chiến thắng trong các tình huống không chiến)
17	Lost Aerial Duels - LostAD	Float	(Thua trong các tình huống không chiến)

Defensive_Profile Dataset			
No.	**Attributes**	**Type**	**Description**
1	Total Tackles - TotTkl	Float	Tổng số pha tắc bóng thành công / 90 phút
2	Dribbled Past - DribPast	Float	Số lần bị đối phương qua người trong tình huống 1v1 / 90’
3	Total Attempted Tackles - TotAttTkl	Float	bao gồm tắc bóng đã thực hiện - bao gồm cả thành công và thất bại / 90’ (= TotTkl + DribPast)
4	Total Interception - TotInt	Float	Số lần đón bóng, cắt đường chuyền mà không cần đối đầu trực tiếp với người cầm bóng / 90’
5	Fouled	Float	Số lần bị phạm lỗi bởi cầu thủ đối phương / 90’
6	Fouls	Float	Số lần phạm lỗi với người khác / 90'
7	Caught Offside - COF	Float	Số lần việt vị / 90'
8	Total Clearance - TotClr	Float	Số phần phá bóng ra khỏi khu vực nguy hiểm / 90’
9	Blocked Shots - BlkSh	Float	(Sút bị chặn): Số lần cú sút của đối phương bị một cầu thủ trong đội phòng ngự chặn lại
10	Blocked Crosses - BlkCr	Float	(Tạt bóng bị chặn): Số lần đường tạt bóng từ biên của đối phương bị chặn lại
11	Blocked Passes - BlkPs	Float	(Đường chuyền bị chặn): Số lần các đường chuyền của đối phương bị chặn lại, ngăn cản bóng đến được đồng đội
12	Total Save - TotSav	Float	
13	OutOfBox Saves - OOBSav	Float	(Cứu thua ngoài vòng cấm): Số lần cứu thua xảy ra ngoài khu vực vòng cấm
14	SixYardBox Saves - SYBSav	Float	(Cứu thua trong hộp 6 yard): Số lần cứu thua xảy ra trong khu vực 6 yard gần khung thành
15	PenaltyArea Saves - PASav	Float	(Cứu thua trong vòng cấm): Số lần cứu thua trong khu vực vòng cấm

Passing_Profile Dataset			
No.	**Attributes**	**Type**	**Description**
Passes - Nhóm dữ liệu về khả năng chuyền bóng			
1	Total Passes - TotPs		Tổng số lần thực hiện các đường chuyền (cả bóng dài và bóng ngắn).
2	Accurate Long Balls - AccLB		Số lượng đường chuyền dài chính xác của cầu thủ.
3	Inaccurate Long Balls - InAccLB		Số lượng đường chuyền dài không chính xác của cầu thủ.
4	Accurate Short Passes - AccSP		Số lượng đường chuyền ngắn chính xác của cầu thủ
5	Inaccurate Short Passes - InAccSP		Số lượng đường chuyền ngắn không chính xác của cầu thủ
Key Pass - Nhóm dữ liệu về các đường chuyền quyết định			
6	Total Key Passes - TotKPs		Tổng số đường chuyền quyết định
7	Long Key Passes - LKPs		Đường chuyền quyết định dài
8	Short Key Passes - SKPs		Đường chuyền quyết định ngắn
Assists - Nhóm dữ liệu về đường chuyền kiến tạo			
9	Total Assists - TotAss		Tổng số đường chuyền kiến tạo
10	Cross Assists - CrAss		Đường chuyền kiến tạo từ quả tạt
11	Corner Assists - CorAss		Đường chuyền kiến tạo từ quả phạt góc
12	Throughball Assists - ThrbAss		Đường chuyền kiến tạo từ đường chuyền xuyên tuyến
13	Freekick Assists - FreAss		Đường chuyền kiến tạo từ quả đá phạt
14	Throw-in Assists - ThrInAss		Đường chuyền kiến tạo từ quả ném biên
15	Other Assists - OthAss		Đường chuyền kiến tạo khác

B. Optimal hyperparameter for compared algorithms

Model	Hyperparameter
Ridge Regression	regularization parameter $\lambda = 0.001$
Support Vector Regressor	Kernel = 'rbf' Gamma (Kernel Coefficient) = 'auto' C (Regularization parameter) = 10 Epsilon ($\varepsilon -$ insensitive Tube) = 0.1
Random Forest Regressor	max_depth = 5 min_samples_leaf = 8 min_samples_splits = 5 n_estimators = 100
XGBoost Regressor	subsample = 0.7 reg_lambda = 5 reg_alpha = 0 n_estimators = 100 min_child_weight = 3 max_depth = 3 learning_rate = 0.1 gamma = 0.2 colsample_bytree = 1.0
LightGBM Regressor	subsample = 0.8 num_leaves = 31 n_estimators = 200 max_depth = 3 learning_rate = 0.05 colsample_bytree = 0.8